
Modern Algebra 1

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Problem 1

(b) Let $\phi : G \rightarrow H$ be a group homomorphism. We wish to show $\phi(g^n) = \phi(g)^n$ for all $n \in \mathbb{Z}$. First, let $n = 0$. Note that $\phi(0) = \phi(e_G)$. By properties of homomorphisms, $\phi(e_G) = e_H$. Furthermore, note that since $\phi(g) \in H$, we have $\phi(g)^0 = e_H$. Therefore, $\phi(g^0) = \phi(g)^0$.

Now suppose that $\phi(g^n) = \phi(g)^n$ for some $n \in \mathbb{N}$. It follows that

$$\phi(g^{n+1}) = \phi(g^n g) = \phi(g^n) \phi(g).$$

By our inductive hypothesis, $\phi(g^n) = \phi(g)^n$. Therefore,

$$\phi(g^n) \phi(g) = \phi(g)^n \phi(g) = \phi(g)^{n+1}.$$

Therefore, the statement holds for $n + 1$. We now have $\phi(g^n) = \phi(g)^n$ for all $n \in \mathbb{N}$.

We must now generalize the proof to $n \in \mathbb{Z}$ by considering negative n . Let $n \in \mathbb{N}$. By definition,

$$g^{-n} = (g^{-1})^n.$$

Since we took $g \in G$ to be an arbitrary element, we have

$$\phi(g^{-n}) = \phi((g^{-1})^n) = \phi(g^{-1})^n.$$

Furthermore, note that by properties of homomorphisms, $\phi(g^{-1}) = \phi(g)^{-1}$. We then have

$$\phi(g^{-1})^n = (\phi(g)^{-1})^n.$$

By definition, this equals $\phi(g)^{-n}$. Therefore, for all $n \in \mathbb{Z}$, $\phi(g^n) = \phi(g)^n$.

After rereading this I realized we actually haven't proven that $\phi(g^{-1}) = \phi(g)^{-1}$, but the proof is simple.

$$e = \phi(gg^{-1}) = \phi(g)\phi(g^{-1}).$$

Therefore, $\phi(g)^{-1} = \phi(g^{-1})$.

Problem 2

Let G be a group with order 99. Notice that $99 = 11 \cdot 3^2$. Let n_3 be the number of Sylow 3-subgroups of G . By Sylow's theorem, $n_3 | 11$ and $n_3 \equiv 1 \pmod{3}$. Since divisors of 11 are only 1 and 11, and since $11 \equiv 2 \pmod{3}$, we know $n_3 = 1$. Now let n_{11} be the number of Sylow 11-subgroups of G . By Sylow's theorem, $n_{11} | 9$ and $n_{11} \equiv 1 \pmod{11}$. The only positive integer $n < 9$ such that $n \equiv 1 \pmod{11}$ is 1, so $n_{11} = 1$.

Let P be the Sylow 3-subgroup with order 9, and let Q be the Sylow 11-subgroup with order 11. By Sylow's theorem, $P, Q \triangleleft G$ since they are the only Sylow 3- and 11-subgroups, respectively. Since they are normal,

$$PQ = \bigcup_{p \in P} pQ = \bigcup_{p \in P} Qp = QP.$$

Therefore, PQ is a subgroup of G . Moreover, $P, Q \leq PQ$, since $e \in P, Q$. By Lagrange's theorem, $|P|, |Q|$ divide $|PQ|$, and since $PQ \leq G$, it follows that $|PQ| \leq |G| = 99$. Note that $|P| = 3^2$ and $|Q| = 11$. Therefore, 9, 11 divide $|PQ|$. We have

$$\text{lcm}(9, 11) = \frac{9 \cdot 11}{\gcd(9, 11)} = \frac{99}{1} = 99.$$

Therefore, $|PQ| \geq 99$ and $|PQ| \leq 99$, and thus $PQ = G$. Therefore, $G \cong P \times Q$.

Since Q is order 11 which is prime, Q is cyclic, and thus abelian. Since P is order 9 which is the square of a prime, by a previous homework problem P is abelian. We thus have $P \times Q$ is abelian, and G is abelian by properties of isomorphisms.

Problem 3

Let $p(x), q(x), r(x) \in R[x]$ such that p, q, r have degrees m, n, ℓ and coefficients a_i, b_i, c_i , respectively.

First, we show $R[x]$ is abelian under addition. By definition,

$$p(x) + q(x) = \sum_{i=0}^{\max\{m,n\}} (a_i + b_i) x^i.$$

Since R is a ring, $a_i + b_i \in R$ for all $a_i, b_i \in R$. It follows that

$$\sum_{i=0}^{\max\{m,n\}} (a_i + b_i) x^i = \sum_{i=0}^{\max\{m,n\}} (b_i + a_i) x^i = q(x) + p(x).$$

Therefore, $R[x]$ is abelian. Moreover, note that $p(x) + q(x)$ is a polynomial with coefficients $(a_i + b_i) \in R$, and thus $p(x) + q(x) \in R[x]$.

Now we show $R[x]$ is associative. Notice that

$$\begin{aligned} (p(x) + q(x)) + r(x) &= \left(\sum_{i=0}^{\max\{m,n\}} (a_i + b_i) x^i \right) + r(x) \\ &= \sum_{i=0}^{\max\{m,n,\ell\}} ((a_i + b_i) + c_i) x^i. \end{aligned}$$

Since R is a ring, addition of coefficients is associative. It follows that

$$\begin{aligned} \sum_{i=0}^{\max\{m,n,\ell\}} ((a_i + b_i) + c_i) x^i &= \sum_{i=0}^{\max\{m,n,\ell\}} (a_i + (b_i + c_i)) x^i \\ &= p(x) + \left(\sum_{i=0}^{\max\{n,\ell\}} (b_i + c_i) x^i \right) \\ &= p(x) + (q(x) + r(x)). \end{aligned}$$

Therefore, $R[x]$ is associative under addition. Notice that $0 \in R[x]$. We have

$$p(x) + 0 = \sum_{i=0}^{\max\{m,0\}} (a_i + 0) x^i = \sum_{i=0}^m a_i x^i = p(x).$$

Therefore, there exists an additive identity. Finally, let $-p(x) \in R[x]$ have coefficients $-a_i$. It follows that

$$p(x) - p(x) = \sum_{i=0}^{\max\{m,n\}} (a_i - a_i) x^i = \sum_{i=0}^m 0x^i = 0.$$

Therefore, $R[x]$ is an abelian group under addition.

Now we show $R[x]$ is closed under multiplication. By definition,

$$p(x)q(x) = \sum_{i=0}^{m+n} \left(\sum_{j=0}^i a_j b_{i-j} \right) x^i.$$

Notice that

$$\sum_{j=0}^i a_j b_{i-j} \in R$$

for all i, j , since R is closed under addition and multiplication. Therefore, the coefficients of this sum are all elements of R , and $p(x)q(x) \in R[x]$.

Now we show that $R[x]$ is commutative. We have

$$p(x)q(x) = \sum_{i=0}^{m+n} \left(\sum_{j=0}^i a_j b_{i-j} \right) x^i.$$

Notice that

$$\sum_{j=0}^i a_j b_{i-j} = a_0 b_i + a_1 b_{i-1} + \cdots + a_{i-1} b_1 + a_i b_0.$$

Since addition and multiplication are commutative over R , this equals

$$= a_i b_0 + a_{i-1} b_1 + \cdots + a_0 b_i = \sum_{j=0}^i a_{i-j} b_j = \sum_{j=0}^i b_j a_{i-j}.$$

Therefore,

$$p(x)q(x) = \sum_{i=0}^{m+n} \left(\sum_{j=0}^i a_j b_{i-j} \right) x^i = \sum_{i=0}^{m+n} \left(\sum_{j=0}^i b_j a_{i-j} \right) x^i = q(x)p(x).$$

Therefore, $R[x]$ is commutative over multiplication. Now we show associativity. We will use the following alternative definition of multiplication:

$$p(x)q(x) = \sum_{i=0}^m \sum_{j=0}^n a_i b_j x^{i+j}.$$

We have

$$\begin{aligned}
p(x)(q(x)r(x)) &= p(x) \left(\sum_{i=0}^n \sum_{j=0}^{\ell} b_i c_j x^{i+j} \right) \\
&= \left(\sum_{i=0}^m a_i x^i \right) \left(\sum_{i=0}^n \sum_{j=0}^{\ell} b_i c_j x^{i+j} \right) \\
&= \sum_{i=0}^m a_i x^i \sum_{j=0}^n \sum_{k=0}^{\ell} b_j c_k x^{j+k} \\
&= \sum_{i=0}^m \sum_{j=0}^n \sum_{k=0}^{\ell} a_i b_j c_k x^{i+j+k}.
\end{aligned}$$

Note that in this last step we apply associativity over R . Applying this property again, we have

$$\begin{aligned}
\sum_{i=0}^m \sum_{j=0}^n \sum_{k=0}^{\ell} (a_i b_j) c_k x^{i+j+k} &= \sum_{i=0}^m \sum_{j=0}^n (a_i b_j x^{i+j}) \left(\sum_{k=0}^{\ell} c_k x^k \right) \\
&= \sum_{i=0}^m \sum_{j=0}^n (a_i b_j x^{i+j}) r(x) \\
&= (p(x)q(x))r(x).
\end{aligned}$$

Therefore, $R[x]$ is associative under multiplication.

Note that the polynomial $1 \in R[x]$, where 1 is the unity in R . Notice that

$$1p(x) = \sum_{k=0}^{m+1} \left(\sum_{l=0}^k 1_l a_{k-l} \right) x^k = \sum_{k=0}^m a_k x^k = p(x).$$

Therefore, 1 is a unity in $R[x]$. Finally, we show distributivity.

$$\begin{aligned}
p(x)(q(x) + r(x)) &= p(x) \left(\sum_{i=0}^{\max\{n,\ell\}} (b_i + c_i) x^i \right) \\
&= \sum_{i=0}^m \sum_{j=0}^{\max\{n,\ell\}} a_i (b_j + c_j) x^{i+j} \\
&= \sum_{i=0}^m \sum_{j=0}^{\max\{n,\ell\}} a_i b_j x^{i+j} + a_i c_j x^{i+j} \\
&= \left(\sum_{i=0}^m \sum_{j=0}^{\max\{n,\ell\}} a_i b_j x^{i+j} \right) + \left(\sum_{i=0}^m \sum_{j=0}^{\max\{n,\ell\}} a_i c_j x^{i+j} \right) \\
&= p(x)q(x) + p(x)r(x).
\end{aligned}$$

In the above proof, we recognized $b_i + c_i$ as the i th coefficient of the polynomial $q(x) + r(x)$, and then used the fact that R was a ring to apply the distributive property. Therefore, R is a commutative ring with unity.

Problem 4

Let R be a ring with $a \in R$ a unit. Since R has a unit, it must have unity since the definition of an inverse requires an identity. Then by definition there exists $a^{-1} \in R$ such that $aa^{-1} = a^{-1}a = 1$, where 1 is the unity. Suppose that $ab = 0$ for some $b \in R$. It follows that $a^{-1}ab = a^{-1}0$. By properties of rings, $a^{-1}0 = 0$. Since a is a unit, $a^{-1}ab = 1b$, and since 1 is the multiplicative identity, $1b = b$. Therefore, $b = 0$. Therefore, a is not a zero divisor.

The converse is not true. Consider the ring $\mathbb{Z}$, which has unity and no zero divisors. However, n has no multiplicative inverse in general, since $1/n \notin \mathbb{Z}$ for $n \neq \pm 1$.

Problem 5

(a) Notice that $2 + i \in S$. Since $\mathbb{C}$ is a ring and $S \subseteq \mathbb{C}$,¹ it follows that

$$(2 + i)(2 + i) = 3 + 4i.$$

Notice that there exists no $n \in \mathbb{Z}$ such that $2n = 3$. Therefore, $3 + 4i \notin S$, and S is not a subring because it is not closed.

(b) Notice that $p(x) = 0 \in S$, and thus $S \neq \emptyset$. We will now demonstrate closure under multiplication. Let $p(x), q(x) \in S$. It follows that $p(3)q(3) = 0 \cdot 0 = 0$. Therefore, $p(x)q(x) \in S$. Finally, let $p(x), q(x) \in R$. We must show $p(3) - q(3) \in R$. It follows that

$$p(3) - q(3) = 0 - 0 = 0.$$

Therefore, $p(3) - q(3) \in R$. By the subring test, S is a subring of $\mathbb{Z}[x]$.

(c) Obviously $S \neq \emptyset$. Let $a + bi, c + di \in S$. Since $\mathbb{H}$ is a ring and $S \subseteq \mathbb{H}$, it follows that

$$(a + bi) - (c + di) = (a - c) + bi - di = (a - c) + (b - d)i.$$

Since $a, b, c, d \in \mathbb{R}$, we have $(a + bi) - (c + di) \in S$. Now notice that

$$(a + bi)(c + di) = ac + adi + bci - bd = (ac - bd) + (ad + bc)i.$$

Since $ac - bd, ad + bc \in \mathbb{R}$, we have $(a + bi)(c + di) \in S$, and by the test for a subring, S is a subring of $\mathbb{H}$.

¹This statement is required to use associativity, commutativity, and distributivity.